

# Jonathan Carroll, PhD

Quantitative Scientist & Research Software Engineer

Adelaide, South Australia | [contact@jcarroll.com.au](mailto:contact@jcarroll.com.au) | +61 407 978 786  
[jcarroll.com.au](http://jcarroll.com.au) | [github.com/jonocarroll](https://github.com/jonocarroll) | ORCID: 0000-0002-1404-5264

## PROFILE

Senior quantitative scientist and research software engineer with 10+ years across multi-modal biomedical research, clinical trial analysis, and fisheries science. Experienced in statistical modelling, data analysis, and production analytics pipelines; author of *Beyond Spreadsheets with R*. Specialist in R with production experience across the full analytical pipeline; from data ingestion and curation through modelling, visualisation, and reporting to non-technical audiences.

Extensive experience with clinical trial data standards (ADaM, SDTM), multi-modal biomedical data integration, and survival and longitudinal analyses. Background in translational immunology, genomics, and large-scale cohort data (UK Biobank). Comfortable operating independently or leading distributed teams in remote environments since 2017.

**Book:** *Beyond Spreadsheets with R*: A beginner's guide to R and RStudio (Manning Publications, 2018, 352 pp.)

## TECHNICAL SKILLS

|                                  |                                                                                                                                                                        |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Statistical methods</b>       | GLMs, GAMs, mixed effects models, survival analysis (Kaplan–Meier, Cox PH), longitudinal data analysis, spatiotemporal modelling, DGE/GSEA, likelihood-based inference |
| <b>Primary language</b>          | R (base and tidyverse, ggplot2, Bioconductor, shiny, targets, dplyr, data.table)                                                                                       |
| <b>Additional languages</b>      | Python, Rust, Haskell, Julia, SQL, Cypher, Bash, C++, JavaScript, APL, Lisp, Scala                                                                                     |
| <b>Data &amp; infrastructure</b> | DuckDB, PostgreSQL, SQLite, Redshift, REST APIs, Docker, Git, CI/CD, UNIX/Linux                                                                                        |
| <b>Clinical data</b>             | ADaM, SDTM, CDISC; FAIR data principles; MultiAssayExperiment                                                                                                          |
| <b>Bioinformatics</b>            | Benchling, Sapio LIMS, UK Biobank, genomics pipelines, antibody discovery infrastructure                                                                               |
| <b>Reporting</b>                 | Quarto, shiny dashboards, peer-reviewed publications, conference presentations                                                                                         |

## INDUSTRY EXPERIENCE

**Bioinformatics Engineer** August 2026 – Current  
*The Kids Research Institute*

- Pipeline generation and maintenance for GWAS and PRS related to lipidomics in indigenous Australian populations

**Senior Research Software Engineer** 2022 – May 2026  
*Human Immunology Biosciences (HIBio) & Biogen*

- Built analytical infrastructure and exploratory dashboards supporting translational science and antibody discovery; connected data across internal systems via REST APIs.
- Migrated a production LIMS from Benchling to Sapio, preserving data integrity and provenance across complex experimental workflows.
- Processed and retrieved petabytes of UK Biobank genotype data for allele-frequency analyses; productionised and extended C++ likelihood code (TrypLik), orchestrating parallel execution via Bash.
- Managed deployment of GoAnywhere secure file transfer for regulated data exchange.

**Research Software Engineer** 2024 – Dec 2025  
*Rancho Biosciences / Roche / Genentech*

- Integrated AI/LLM tooling into R shiny dashboards for exploratory clinical data analysis; prompt engineering for biomedical summarisation tasks.
- Data processing and analysis in Redshift; exploration of ADaM and SDTM clinical trial datasets.

**Technical Lead / co-Product Owner** 2021 – 2022  
*Cancer Immunology, Genentech / Roche*

- Led development of a large-scale interactive genomics visualisation platform (R shiny) for life scientists without coding experience; coordinated a distributed team of 10 developers across multiple time zones.
- Platform incorporated DGE, GSEA, Kaplan–Meier survival analyses, volcano plots, t-tests, and laboratory result trend visualisations for clinical trial cohorts.
- Managed agile delivery, devOps, CI/CD pipelines, database design (SQLite/SQL), and Docker-based deployment.

**Data Curator / Integrator & Software Developer** 2017 – 2022  
*Cancer Immunology, Genentech / Roche*

- Curated and integrated multi-modal high-dimensional data (genomics, proteomics, clinical trial outcomes) from

ADaM/SDTM-structured datasets into a FAIRified internal DataMart (MultiAssayExperiment).

- Performed Kaplan–Meier and Cox proportional hazards survival analyses for clinical trial cohorts; longitudinal modelling of concomitant medications and laboratory results across trial timepoints.
- Conducted in silico analyses linking multi-omics data to patient outcomes; contributed analyses to peer-reviewed publications (Nature Communications, PNAS, medRxiv).
- Developed R/Bioconductor tooling and shiny applications; authored reusable packages supporting downstream analytical teams.

## Senior Research Officer / Fisheries Modeller

2013 – 2017

*South Australian Research and Development Institute (SARDI)*

- Developed and maintained stock assessment models for South Australian fisheries using GLMs, GAMs, mixed effects models, and custom likelihood-based approaches in C++ (ADMB).
- Conducted spatiotemporal catch rate analyses and GIS analyses to characterise population dynamics under environmental covariates (wind, swell).
- Produced annual statistical reports and peer-reviewed publications; provided statistically rigorous briefings with uncertainty quantification to government policymakers to inform quota-setting decisions.

## RESEARCH EXPERIENCE

### Postdoctoral Research Fellow

2009 – 2013

*Special Research Centre for the Subatomic Structure of Matter, University of Adelaide*

- Dense matter nuclear astrophysics: equation-of-state modelling for hybrid neutron stars via numerical simulation in Fortran with MPI parallelisation.
- Muonic hydrogen spectroscopy and the proton radius puzzle; nonperturbative relativistic quantum mechanical calculations.

## EDUCATION

### PhD, Theoretical Physics

2006 – 2009

*University of Adelaide*

Thesis: *Applications of the Octet Baryon Quark-Meson Coupling Model to Hybrid Stars* — arXiv:1001.4318

### Bachelor of Science (Honours), Physics & Theoretical Physics

2002 – 2005

*University of Adelaide*

## SELECTED PUBLICATIONS

### Book

Carroll J. (2018). *Beyond Spreadsheets with R*. Manning Publications, 352 pp.

### Peer-reviewed articles (selected)

Negative impact of the GABA pathway on PD-1/PD-L1 immunotherapy. *Submitted 2025*.

Natural killer cell educating KIR/HLA combinations impact survival in anti-PD-L1 treated cancer patients. *medRxiv*, 2022.

Genetic variation associated with thyroid autoimmunity shapes the systemic immune response to PD-1 checkpoint blockade. *Nature Communications* 12, 3355 (2021).

Polygenic risk for skin autoimmunity impacts immune checkpoint blockade in bladder cancer. *PNAS* 117(22), 12288–12294 (2020).

Influence of wind and swell on catch rates in a dive fishery. *Journal of Shellfish Research* 35(3), 685–694 (2016).

Nonperturbative relativistic calculation of the muonic hydrogen spectrum. *Physical Review A* 84(1) (2011).

## SELECTED OPEN-SOURCE SOFTWARE

**Published R packages** (author & maintainer unless noted): {ggeasy} · {charcuterie} · {ntfy} · {mathpix} · {DFplyr} (Bioconductor) · {benchlingapi} · {weatherOz} (contributor) · {datapasta} (contributor)

## COMMUNITY & COMMUNICATION

- **Blog:** Regular technical writing at jcarroll.com.au covering R, data analysis, and multi-language programming explorations (R, Rust, Haskell, Julia, APL, and others).
- **rweekly.org:** Editor — compilation, curation, and editing of the weekly R community newsletter (2016 – present).
- **Conference presentations:** DataEngBytes and other data science venues.
- **Governing Council:** Highbury Preschool / Highbury Primary / Modbury High (Chair, Vice Chair; 2016 – present).
- **Chess Club Teacher** (volunteer): Highbury Primary School (2022 – present).

